

1.

$$\text{Ukuran} = 8 = 96$$

$$\text{bit} = 12 \text{ byte}$$

$$\text{ASCII} = 3 \text{ bit}$$

2.

$$1. a. (6/12), (2/12), d (2/12), j (1/12), s (1/12)$$

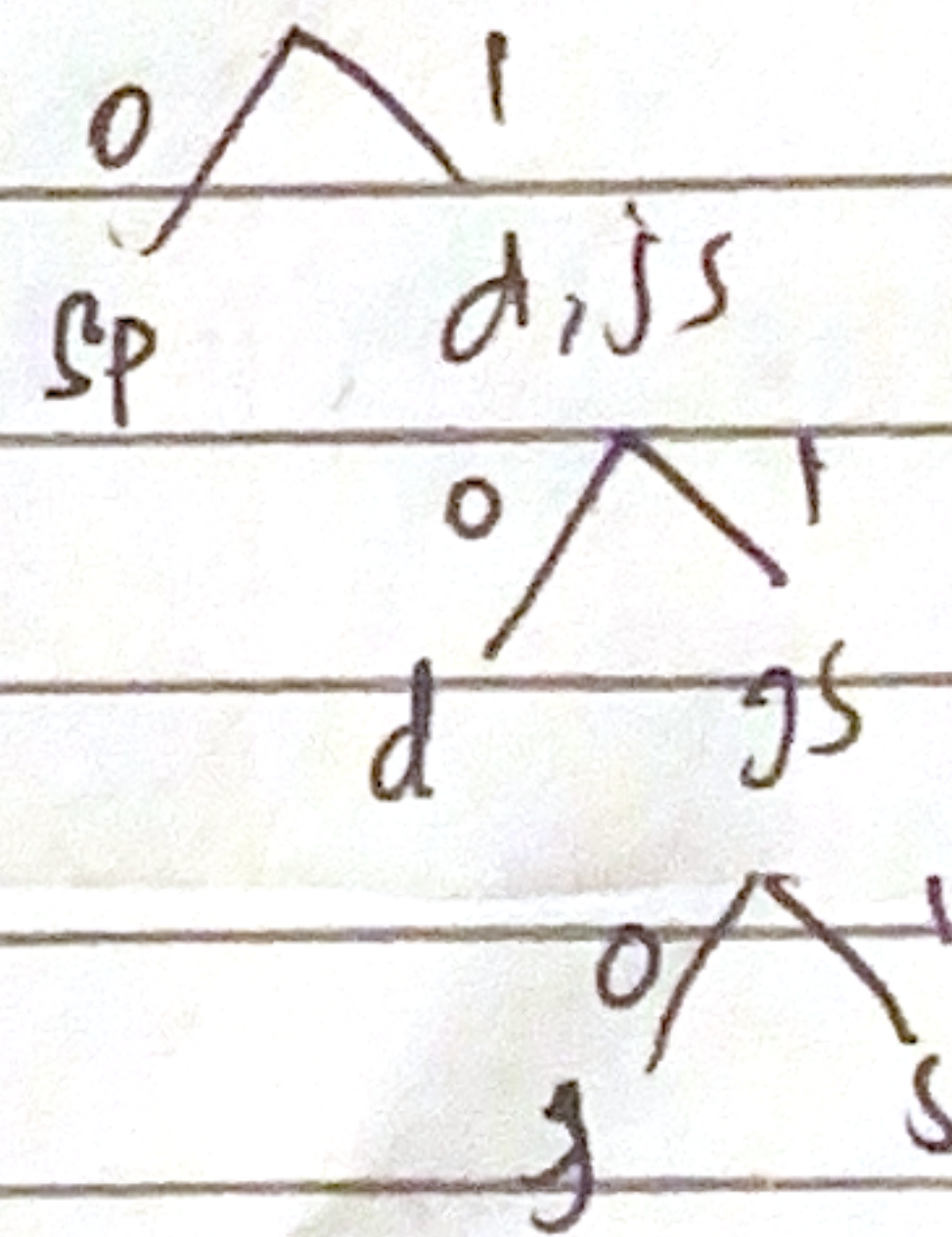
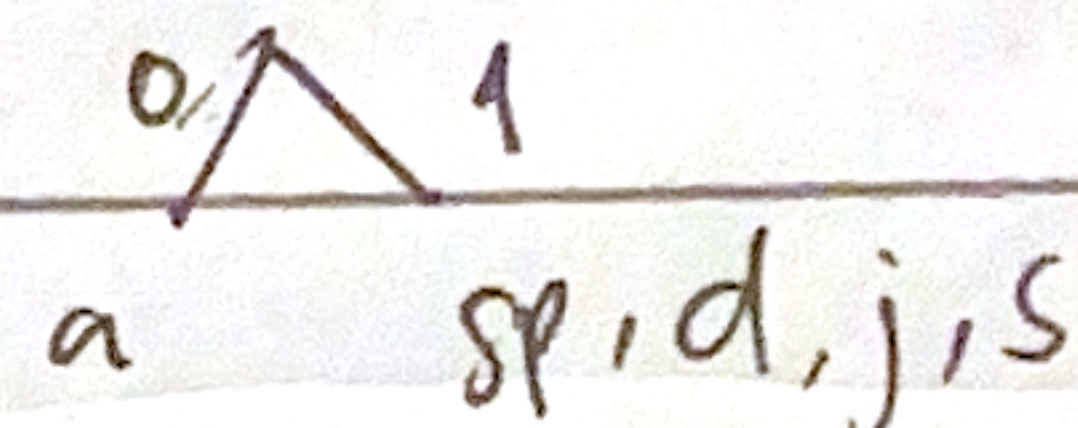
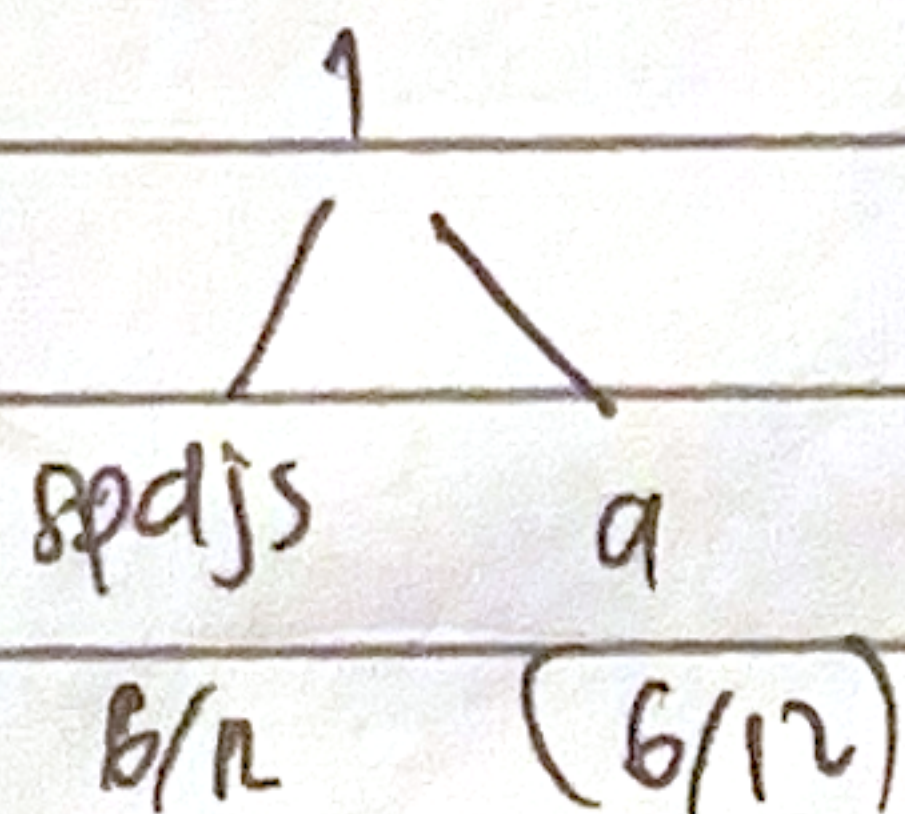
$$2. a (6/12), (2/12), d (2/12), js (2/12)$$

$$3. a (6/12), (2/12), djs (4/12)$$

$$4. a (6/12), d (6/12)$$

$$5. a djs sp (12/12) = (1)$$

a, sp, d, j, s



Kode Huffman

011001001100101111011100

3.

Average

$$= 0,5 + 0,5 + 0,333 + 0,333 + 0,333$$

$$1,91 \text{ bit/karakter} = 2$$

4.

redundansi = Avg - entropi

$$\text{Entropy} = \sum p_i \times \log \frac{1}{p_i}$$

$$a = 6/12 \times 2 \log 2 = 0,5$$

$$d = 2/12 \times 2 \log 2 = 0,43$$

$$g = 1/12 \times 2 \log 2 = 0,13$$

$$j = 1/12 \times 2 \log 2 = 0,13$$

$$g_p = 2/12 \times 2 \log 2 = 0,43$$

$$\text{redundância} = 2 - 1,96$$

$$= 0,04$$